

1. 2012Q20

Which of the following descriptions about the function of the liver is correct?

- A. detects blood glucose level
- B. breaks down red blood cells
- C. secretes glucagon into the blood
- D. secretes enzymes into the small intestine

2. 2012Q34

After consuming a boiled egg, chemical digestion begins in the

- A. mouth.
- B. oesophagus.
- C. stomach.
- D. small intestine.

3. 2013Q26

Which of the following nutrients enter the epithelial cells of the small intestine mainly by simple diffusion?

- A. amino acids
- B. fatty acids
- C. monosaccharides
- D. nucleotides

4. 2014Q1

Which of the following is a catabolic process?

- A. Conversion of glucose to glycogen
- B. Absorption of glucose
- C. Emulsification of fat
- D. Digestion of starch

5. 2014Q24

Which of the following dental formulae best represents the dentition shown in the photograph of the X-ray?



A.
$$\frac{2123}{2123}$$

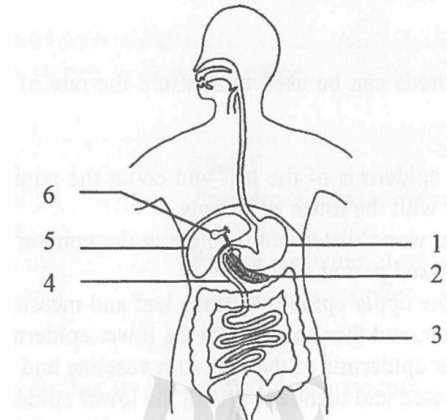
B.
$$\frac{2132}{2132}$$

C.
$$\frac{3212}{3212}$$

D.
$$\frac{2312}{2312}$$

6. 2014Q25

Questions 25 and 26 refer to the diagram below, which shows the human digestive system:



Physical digestion takes place at

- A. 1 and 3.
- B. 1 and 4.
- C. 3 and 4.
- D. 1, 3 and 4.

7. 2014Q26

Which of the following structures are responsible for producing digestive juices that help the digestion of fat?

- A. 2 and 5
- B. 2 and 6
- C. 5 and 6
- D. 2, 5 and 6

8. 2015Q7

Which of the following combinations correctly matches the gland, the enzyme secreted and the optimum pH of the enzyme?

	<i>Gland</i>	<i>Enzyme</i>	<i>Optimum pH</i>
A.	gastric gland	carbohydrase	2
B.	liver	lipase	8
C.	salivary gland	amylase	11
D.	pancreas	protease	11

9. 2015Q8

After eating a hamburger, chemical digestion begins in the

- A. mouth cavity.
- B. oesophagus.
- C. stomach.
- D. small intestine.

10. 2015Q9

Which of the following combinations correctly describes the absorption of water in the alimentary canal?

- | | Occurs mostly in | Major reason |
|----|-------------------------|---|
| A. | ileum | it is the longest part of the digestive tract |
| B. | ileum | most digested food is absorbed in this region |
| C. | large intestine | its function is water absorption |
| D. | large intestine | absorption of food has completed in this region |

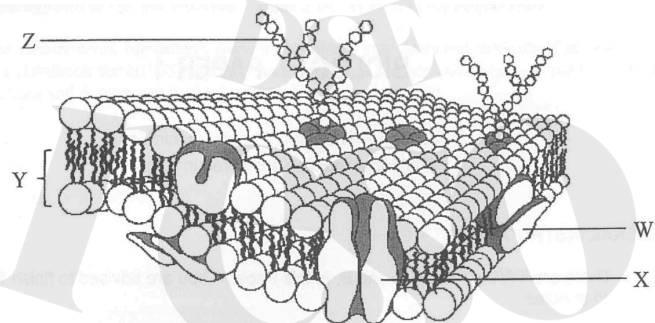
11. 2015Q10

After absorption in small intestine, most fat is first transported to the

- large intestine.
- pancreas.
- heart.
- liver.

12. 2016Q2

Question 2 refers to the schematic diagram below, which shows the structure of a cell membrane:



If the membrane is located at the inner surface of the small intestine, the absorption of amino acids is likely to be achieved via

- W.
- X.
- Y.
- Z.

13. 2016Q3

In which of the following pairs of carbohydrates can Benedict's test be used to distinguish the two carbohydrates from one another?

- sucrose and starch
 - sucrose and maltose
 - glucose and maltose
 - glucose and starch
- (1) and (3) only
 - (1) and (4) only
 - (2) and (3) only
 - (2) and (4) only

14. 2016Q5

Questions 5 and 6 refer to the diagram below, which shows the nutrition label of a food product:

Nutrition Information	Per serving
Energy	62 Kcal
Protein	3.1 g
Total fat	3.5 g
- Saturated fat	2.4 g
- Trans fat	0 g
Total carbohydrate	4.6 g
- Dietary fibre	0 g
- Sugars	4.6 g
Sodium	44 mg
Calcium	110 mg

Which of the following food substances provides most of the energy content in this food product?

- fat
- protein
- sodium
- carbohydrate

15. 2016Q6

The food product bearing this nutrition label is most likely to be

- milk.
- bread.
- sausages.
- potato chips.

16. 2016Q7

Which of the following organs produces digestive juice that **does not** contain enzymes?

- liver
- mouth
- stomach
- pancreas

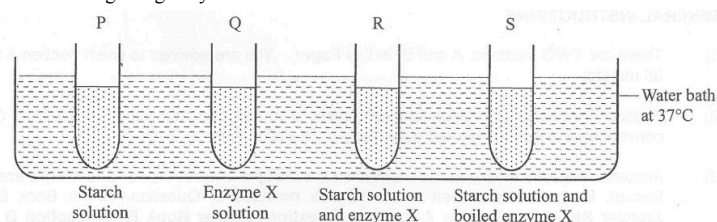
17. 2017Q1

Which of the following combinations correctly matches the process with the type of metabolism involved?

- | Process | Type of metabolism |
|---|---------------------------|
| A. digestion of lipids in the small intestine | anabolism |
| B. storage of excess energy in the form of glycogen | catabolism |
| C. assimilation of amino acids to form muscle fibres | anabolism |
| D. absorption of digested food in the small intestine | catabolism |

18. 2017Q2

Questions 2 to 4 refer to the diagram below, which shows four test tubes prepared by a student to investigate the action of a starch-digesting enzyme X:



In which of the following regions of the human alimentary canal can enzyme X be found?

- (1) mouth cavity
 - (2) stomach
 - (3) small intestine
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

19. 2017Q3

The student conducted some tests on the content of each test tube at the beginning and after 30 minutes.

Which of the following correctly shows the results of the tests for tube R at the beginning?

	Benedict's test	Iodine test	Test for proteins
A.	negative	positive	positive
B.	negative	positive	negative
C.	positive	negative	negative
D.	positive	negative	positive

20. 2017Q4

Which of the following is **not** the purpose of the experimental design?

	Design	Purpose
A.	setting up tube P	to show the result of iodine test if starch is present
B.	setting up tube Q	to show that enzyme X alone cannot give positive result in Benedict's test
C.	setting up tube S	to show that enzyme X is denatured after boiling
D.	setting up water bath	to simulate the temperature of the human body

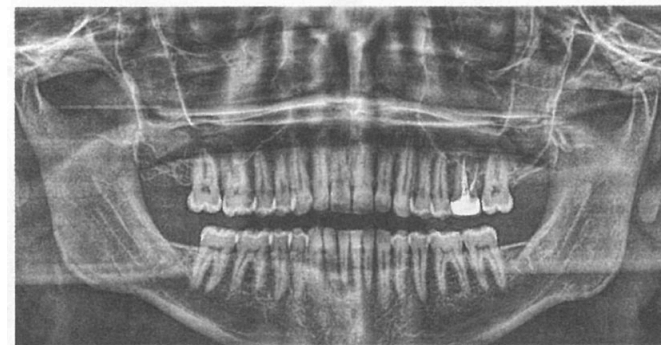
21. 2018Q18

Some people claim that we should chew food for a longer time before swallowing. Which of the following statements about this claim is **incorrect**?

- A. This stimulates the secretion of saliva.
- B. This moistens the food for easier swallowing.
- C. This provides enough time for digestion of starch into glucose.
- D. This increases the surface area of the food for chemical digestion.

22. 2018Q23

The following picture shows an X-ray photograph of the dentition of a person:

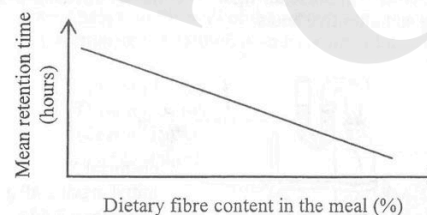


Which of the following dental formulae correctly describes the dentition in the above photograph?

- A. $\frac{2212}{2212}$
- B. $\frac{2122}{2122}$
- C. $\frac{2131}{2131}$
- D. $\frac{2113}{2113}$

23. 2018Q25

The graph below shows the relationship between the dietary fibre content of a meal and the mean retention time (i.e. the duration for which the undigested materials stay in the large intestine) in the human body:

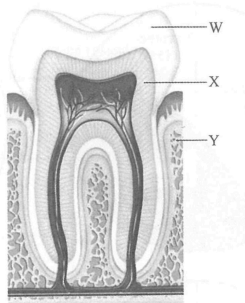


Which of the following can be deduced from the graph?

- A. Meals with more dietary fibre can increase the bulk of the faeces.
- B. Meals with more dietary fibre can hold more water, and so soften faeces better.
- C. Meals with less dietary fibre have more nutrients and a longer time is required for complete absorption.
- D. Meals with less dietary fibre result in harder faeces due to an increased time for water absorption.

24. 2019Q23

Questions 23 and 24 refer to the diagram below, which shows a section of one type of tooth and its associated structures:



Which of the following are living tissues that contain a large amount of calcium salt?

- A. W and X
- B. W and Y
- C. X and Y
- D. W, X, and Y

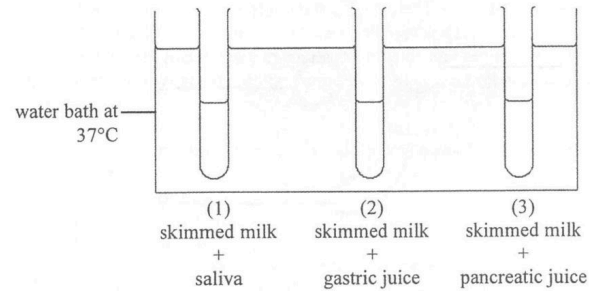
25. 2019Q24

The number of this type of tooth in the milk dentition is

- A. 0.
- B. 4.
- C. 8.
- D. 12.

26. 2019Q25

The diagram below shows a water bath with three test-tubes containing different mixtures:

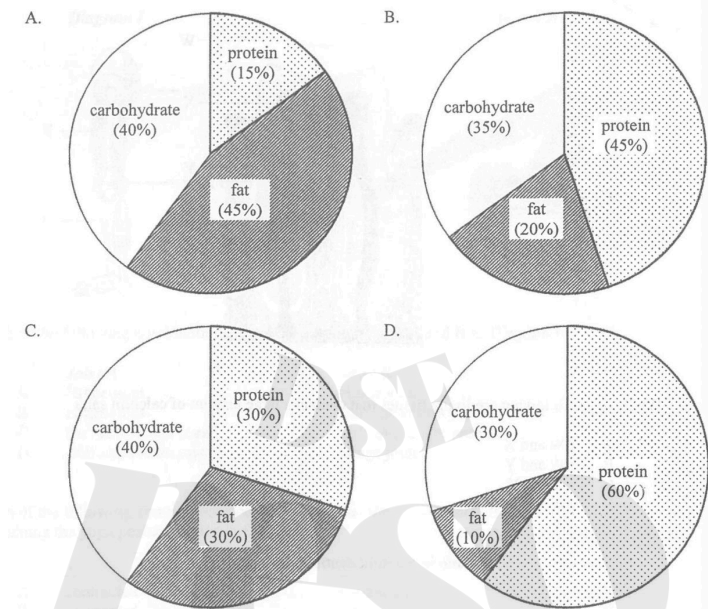


Chemical digestion of food takes place in test-tubes

- A. (1) and (2) only.
- B. (1) and (3) only.
- C. (2) and (3) only.
- D. (1), (2), and (3).

27. 2019Q26

The following charts show the composition of four different foodstuffs. Which foodstuff yields the highest amount of energy per gram?



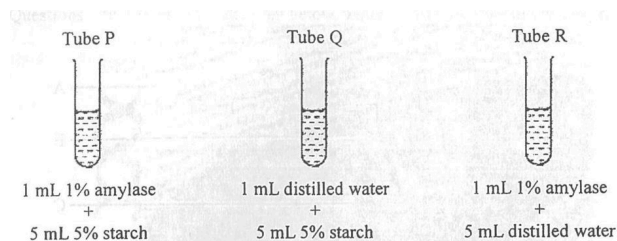
28. 2021Q7

Which of the following correctly describe the absorption of amino acids in the small intestine?

- (1) Amino acids are absorbed into lacteals.
 - (2) The absorption is assisted by membrane proteins.
 - (3) Amino acids can move along or against a concentration gradient.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2), and (3)

29. 2021Q9

Questions 9 and 10 refer to the diagram below, which shows three set-ups used in the study of the activity of amylase:



We can conclude the action of amylase on starch by comparing the results of tubes

- A. P and Q only.
- B. P and R only.
- C. Q and R only.
- D. P, Q, and R.

30. 2021Q10

If the following tests are conducted on the reaction mixture of tube P, which test(s) will give the same result at the beginning of the experiment and after 10 minutes?

- (1) iodine test
- (2) albutix paper
- (3) Benedict's test
- A. (1) only
- B. (2) only
- C. (1) and (2) only
- D. (2) and (3) only

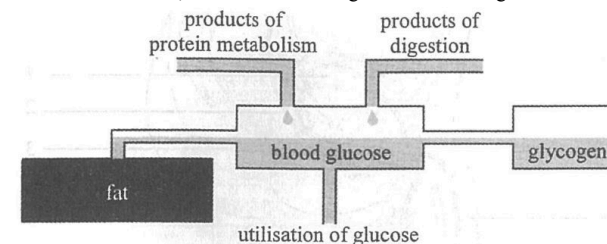
31. 2021Q19

Which of the following segments of the alimentary canal absorbs the largest amount of water?

- A. the oesophagus
- B. the stomach
- C. the small intestine
- D. the large intestine

32. 2021Q27

Question 27 refers to the model below, which shows the regulation of blood glucose level in humans:

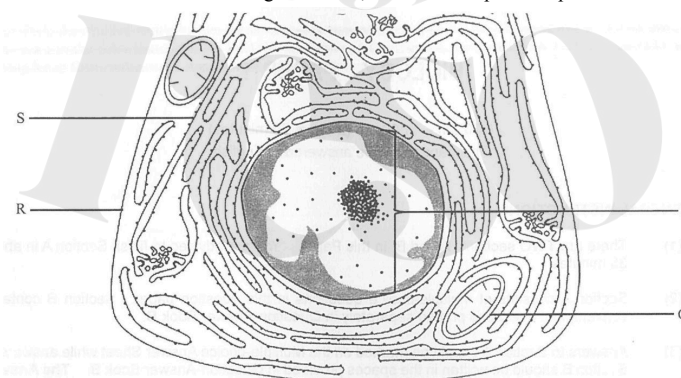


When the blood glucose level is higher than normal, which of the following changes is *incorrect*?

- A. The utilisation of glucose will increase.
- B. The conversion of fat to glycogen will increase.
- C. The conversion of blood glucose to fat will increase.
- D. The conversion of blood glucose to glycogen will increase.

33. 2023Q2

Questions 2 and 3 refer to the schematic diagram below, which shows part of a pancreas cell:



If the pancreas cell is supplied with radioactively labelled amino acids, which of the following cellular components is likely to have an increase in radioactivity first?

- A. P
- B. Q
- C. R
- D. S

34. 2023Q3

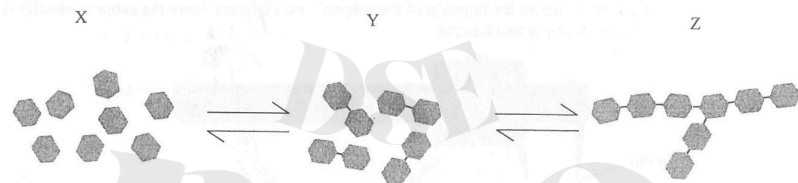
The radioactivity is later detected in the content of the small intestine. Which of the following substances is / are responsible for the presence of the radioactivity?

- (1) protease
- (2) glucagon
- (3) sodium hydrogencarbonate
- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

Which of the following features of the small intestine allow it to be the major site of food absorption?

36. 2024O20

Key:  six-carbon sugars



- (1) Conversion of Z to Y takes place in the stomach.
- (2) Conversion of X to Z takes place in the liver.
- (3) Z is present in muscle.

- A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2), and (3)

Patients who have had their gall bladder removed should eat less oily food. This is because they are **unable** to:

- A. store bile.
- B. store lipase.
- C. produce bile.
- D. produce lipase.

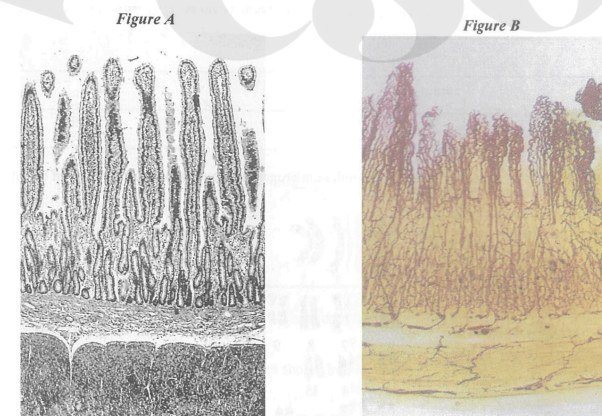
1. 2012Q10

In mammals, the production of bile salts is very limited at birth and during early developmental stages. In an investigation of the effect of bile supplementation on fat digestion in piglets, four groups of piglets were fed with the same diet except that bile was added to diets in the proportion of 0%, 0.15%, 0.30%, and 0.45% respectively for 15 days. Faecal samples were collected each day for analysis and the body weights of the piglets were monitored. The results are shown in the table below:

	Bile level in diet (%)			
	0	0.15	0.3	0.45
Average initial body weight (kg)	4.90	4.46	4.65	4.52
Average final body weight (kg)	7.15	7.36	7.81	7.92
Average dry weight of fat in the faeces (%)	6.52	5.81	4.65	3.78

- How do bile salts help with fat digestion? (2 marks)
- What was the effect of bile supplementation on fat digestion in the above investigation? Explain your answer. (2 marks)
- Suggest why it is important to monitor the body weight of the piglets in this investigation. (1 mark)
- Suggest an *in vitro* experimental method that allows a direct measurement of the digestion of fat in a laboratory. (3 marks)

Figure A shows a section of part of the human alimentary canal. Figure B shows another section of the same part with blood vessels stained.



- a) Which part of the alimentary canal is shown in the figures? Support your answer with evidence. (2 marks)
- b) For each of the above figures, describe one observable feature and explain how it is related to the functioning of this part of the alimentary canal.
- Figure A: (2 marks)
- Figure B: (3 marks)

3. 2014Q11 - Essay

Recently, some people have adopted a diet rich in lean meat in order to lose weight and build muscle. They may be able to achieve these aims but there are some health problems associated with this diet. Discuss the pros and cons of this controversial diet with regard to the nutritional needs of our body and protein metabolism. (12 marks)

4. 2015Q7

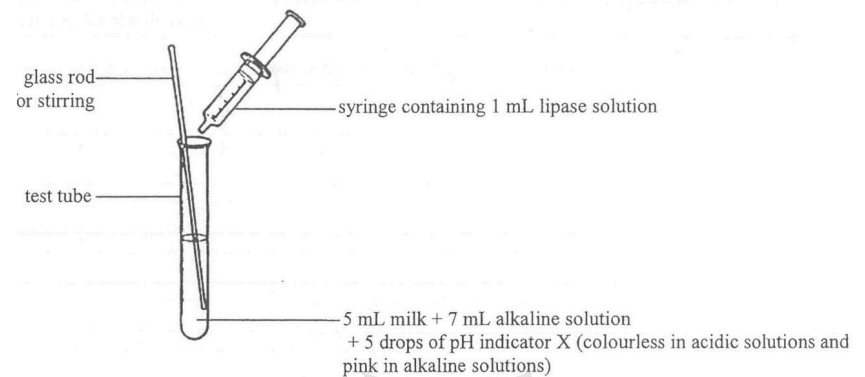
Hydrogen peroxide is a by-product of some metabolic reactions inside our body. If it were allowed to accumulate, it would kill our cells. Fortunately, we have an enzyme called catalase which speeds up the breakdown of hydrogen peroxide into water and oxygen. A student planned to investigate the effect of temperature on the activity of catalase. Below are the steps the student has drafted for his investigation:

- (1) Extract catalase from an animal organ.
 - (2) Mix 5 mL 0.1% hydrogen peroxide solution with 1 mL catalase extract.
 - (3) Place the mixture in a water bath set at 0°C.
 - (4) Measure and record the volume of oxygen gas released in the first 5 minutes.
 - (5) Repeat steps (2) to (4) with the water bath set at 20°C, 40°C, and 60°C.
- a) Suggest an animal organ in which catalase is present in great abundance and from which the enzyme can be obtained. Explain why this organ has so much catalase. (2 marks)
- b) The student has missed out an important step in his drafted procedure. What is it? Explain the importance of this step. (2 marks)
- c) You are provided with the following apparatus and materials: (3 marks)
measuring cylinder, boiling tube, one-hole stopper, glass tubing, rubber tubing, pipette, ink, water trough, clip
- Choose the appropriate apparatus and materials to assemble a set-up for measuring the rate of oxygen production. Draw the set-up in the space below.

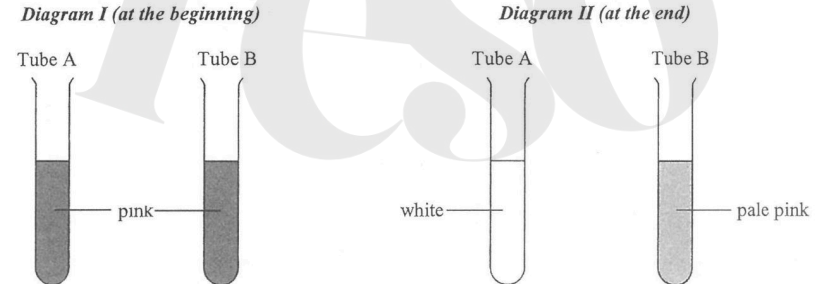
Set up for measuring the rate of oxygen production

5. 2018Q8

A student used the following set-up to compare the fat digestion of full fat fresh milk and half fat fresh milk:



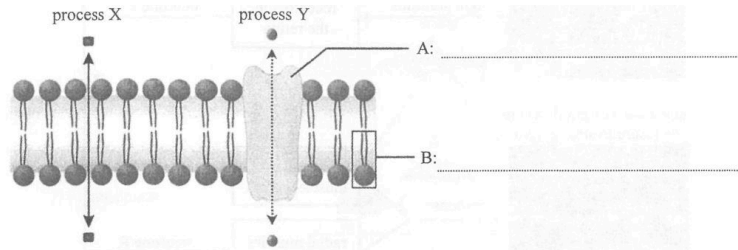
- a) Write a simple word equation of fat digestion. (2 marks)
- b) Two test tubes, each with a different type of milk, were prepared. The colour of each mixture was recorded immediately after the addition of lipase solution, and then monitored until there was no more change in colour. The colours of the mixtures at the beginning and the end are shown in Diagram I and Diagram II respectively.



- i) What is the independent variable in this investigation? (1 mark)
- ii) Describe the results of this investigation. (2 marks)
- iii) Which test tube (A or B) was the one with full fat fresh milk? Explain your answer. (4 marks)

6. 2020Q4bi, bii

The following diagram shows a cell membrane and the movement of some substances across it:



- b) X and Y represent two different processes by which substances pass through the cell membrane.
- Give **one** digested product which is absorbed through process X in the human body. (1 mark)
 - Give **one** digested product which is absorbed through process Y in the human body. (1 mark)

7. 2021Q1

The diagram below shows the presence of gallstones in some part of the human digestive system:



- Label structure A. (1 mark)
- With reference to **two** components of the secretion released from duct B, explain how the condition shown in the above diagram would lead to a decrease in rate of fat digestion. (4 marks)

8. 2022Q1

The following are some life processes in humans:

- Egestion (A)
Excretion (B)
Feeding (C)
Growth (D)
Respiration (E)

- Use the letters to construct an equation showing the relationship of these processes in energy flow. (1 mark)

$$\boxed{} = \boxed{} - \boxed{} - \boxed{} - \boxed{}$$

- In terms of the energy value of each process, explain why the D/C ratio for vegetarians is lower than that for non-vegetarians. (3 marks)

9. 2022Q11 - Essay

Recently, the use of the ketogenic diet for achieving weight loss is becoming popular. In fact, this high-fat, moderate-protein, and very-low-carbohydrate diet has been used as an approach to control the blood glucose level in diabetics. However, the effectiveness of this diet in achieving weight loss is still controversial.

Describe how a ketogenic diet can be used to control the blood glucose level in diabetics. Evaluate the possibility of using this diet for weight loss and discuss the health concerns of adopting such a diet for healthy persons. (12 marks)

10. 2023Q10a, bi, bii1, bii2

Cassava is a crop which grows in areas with poor soil and a low rainfall. It produces starchy root tubers which serve as a major food source in Africa.

- Give the location(s) where the chemical digestion of starch takes place in the human digestive tract. (1 mark)
- Table I below shows some nutritional information of cassava while Table II lists the daily energy and protein requirements recommended for boys at age 16:

Table I		Table II	
Fresh weight (g) from which 100 g dry weight is yielded	250		Daily requirement
Energy (kJ per 100 g dry weight)	2 675	Energy (kJ)	11 100
Protein (g per 100 g dry weight)	3.5	Protein (g)	52

In Africa, some low-income families may rely only on cassava for food for a long period.

- A 16-year-old boy relies only on cassava for food. Calculate the fresh weight of cassava he needs to consume so as to meet the recommended daily energy requirement. (1 mark)
- After consuming cassava only for a period of time, this boy develops swollen feet due to the accumulation of tissue fluid.
 - How much protein can be obtained from the amount of cassava consumed in a (i)? (1 mark)
 - According to Table II, predict the difference of the blood protein level of this boy when compared with that of normal healthy boys of the same age. Explain your answer. (2 mark)

11. 2024Q6

Shirley came across an article about some beans containing an amylase inhibitor as a defense against insects. She wondered if the amylase inhibitor would also work in the human body and if it did, whether it could be used as a food supplement for weight management. She discussed the idea with her classmate Johnson. They had different ideas:

Shirley: I think we should test if the bean extract can inhibit pancreatic amylase.

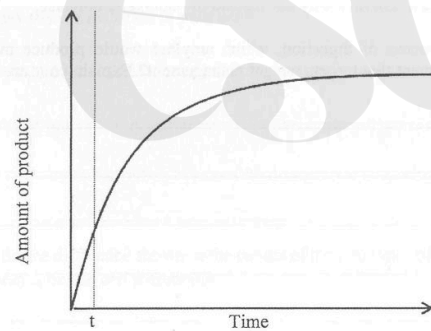
Johnson: Perhaps we can use salivary amylase instead of pancreatic amylase.

- a) With reference to the process of digestion, which amylase would produce more valid results for developing a food supplement that targets weight management? Explain your answer. (3 marks)

- b) The table below shows the reaction mixtures prepared for the investigation:

Solution	Volume of solution used in each set-up (mL)	
	Set-up I	Set-up II
1% starch solution	15	15
Amylase solution	5	5
Bean extract	0	5
Buffer solution (to maintain the pH)	5	5
Water	5	0

- i) Explain the purpose of adding water to set-up I. (2 marks)
- ii) Suggest **one** method to determine the rate of starch digestion and state clearly the measurement taken to show the rate of starch digestion. (2 marks)
- c) The graph below shows the amount of product formed over time when amylase is working normally:



The experiment was repeated with the addition of bean extract at time t. If the bean extract can inhibit the amylase being studied, what will be the change in the amount of product formed? On the above graph, sketch a line to show the results. (1 mark)

- d) Shirley and Johnson shared their ideas with their professor. Their professor suggested that they should conduct an *in vivo* experiment using mice with the control group fed with starchy food and the experimental group fed with a mixture of starchy food and bean extract.
- i) Explain why the result of an *in vivo* experiment is more valid than that of *in vitro* experiment in this case. (1 mark)
- ii) Apart from monitoring the change in body weight of the mice, their professor suggested that they should take blood samples from the mice after the meals for analysis. Which component of the blood should they monitor? What would be the expected results of the control group and the experimental group if their ideas actually worked? (2 marks)
- e) Suggest how the amylase inhibitor helps the bean defend against insects. (1 mark)

Ans

2012Q20: B (64%)	2012Q34: C (747%)	2013Q26: B (33%)	2014Q1: D (64%)
2014Q24: A (63%)	2014Q25: B (47%)	2014Q26: A (62%)	2015Q7: D (47%)
2015Q8: A (76%)	2015Q9: B (36%)	2015Q10: C (43%)	2016Q2: B (69%)
2016Q3: D (43%)	2016Q5: A (52%)	2016Q6: A (68%)	2016Q7: A (80%)
2017Q1: C (67%)	2017Q2: B (67%)	2017Q3: A (34%)	2017Q4: C (23%)
2018Q18: C (63%)	2018Q23: B (74%)	2018Q25: D (36%)	2019Q23: C (45%)
2019Q24: C (41%)	2019Q25: C (68%)	2019Q26: A (80%)	2021Q7: C (65%)
2021Q9: A (57%)	2021Q10: B (52%)	2021Q19: C (73%)	2021Q27: B (75%)
2023Q2: D (60%)	2023Q3: A (56%)	2023Q18: A (54%)	2024Q20: C (52%)
2024Q21: A (78%)			

2012Q10

- 10 (a) • bile salt emulsify fat into droplets (1)
• such that there is an increase in surface area for the action of lipase / enzyme (1)
- (b) • as the bile supplementation increased, the fat content of the faeces decreased (1)
• this indicates increased digestion of fat (1)
- (c) • to show that the addition of bile supplementation does not adversely affect the growth of the pigs (1) / indicate the effectiveness of the bile supplementation on promoting pigs' growth / effectiveness of fat absorption (1)
- (d) Concept for mark award:
(1): suitable substrates and correct enzyme used in the experiment (1)
(2): identification of parameter for measuring the dependent variable (1)
(3): expected results was provided (1)
- e.g. • prepare an alkalized mixture of lipase, bile salts, and oil (1)
• add pH indicator into the mixture / use a data logger with pH sensor / pH meter to show the change in pH of the mixture (1)
• the faster the drop in the pH of the mixture, the faster the digestion of fat (1)
Accept other reasonable experiments

8 marks

2013Q3

- 3 (a) • it is taken from the small intestine / ileum (1)
• as evidenced by the presence of villi in the region (1)
- (b) Figure A: finger-like projections (villi) (1) provide a larger surface area for the absorption of digested food substances (1)
- Figure B: rich supply of blood / dense capillary network (1) helps transport away the absorbed substances quickly (1), this maintains a steep concentration gradient of these food substances to facilitate the absorption (1)

7 marks

2014Q11

11 Lose weight and build muscle (max. 4 marks)

- lean meat does not contain much fat and carbohydrates; this reduces the energy intake (1)
- when the energy intake is lower than the energy expenditure (1), our body will utilize food reserve, body fat in this case, to support our daily activities (1)
- protein in the lean meat will be digested to form amino acids (1)
- amino acids will be assimilated to form muscle fibers (1) according to the needs of the body

Health problems associated with the unbalanced diet (max. 2 marks)

- such diet may lack other essential nutrients which are important for our health (1) such as certain minerals and vitamins
- insufficient intake of minerals and vitamins leads to deficiency diseases (1) (may cite specific examples)

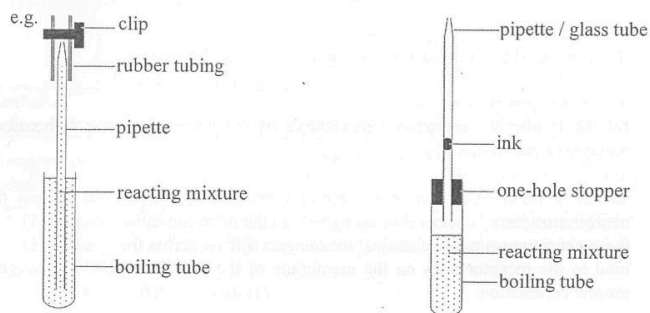
Health problems associated with protein metabolism (max. 3 marks)

- excess amino acids will be deaminated in the liver forming urea (1)
- and the urea will be excreted through the kidney (1)
- this creates heavy workload to both the liver and the kidney (1) and may lead to failure of their functioning (1)

C=max. 3
12 marks

2015Q7

- 7 (a) • liver (1)
• it is the organ where many metabolic reactions take place / it is the organ for detoxification (1) (2)
- (b) • the hydrogen peroxide and catalase solution should be incubated at the set temperature for 10 minutes before mixing (1)
• to ensure that the mixture has reached the set temperature at the beginning of the reaction (1) (2)
- (c) the set-up can collect gas (1)
the set-up can measure the volume of the gas (1)
labels (1)



(3)

2018Q8

- 8 (a) • Triglyceride / fat $\xrightarrow{\text{lipase}}$ glycerol + fatty acids (2)
- (b) (i) • type of milk (1) (1)
- (ii) • colour of the mixture in Tube A turned from pink to white (1)
• colour of the mixture in Tube B turned from pink to pale pink / became pale (1) (2)
- (iii) • Tube A (1)
• Tube A turned white, showing that the mixture in Tube A was acidic (1)
• Tube B remained pale pink, showing that the mixture in Tube B was alkaline (1)
• full fat fresh milk releases more fatty acid than half fat fresh milk after fat digestion (1), eventually resulting in the acidic condition of the mixture in Tube A (4)

9 marks

2020Q4bi, bii

- 4 (b) (i) • glycerol / fatty acid (any **one**) (1) (1)
- (b) (ii) • glucose / simple sugar / amino acid (any **one**) (1) (1)

2021Q1

- 1 (a) pancreas (1) (1)
- (b) Concept for mark award:
• identification of the components in the secretion of duct B which are related to fat digestion (1) and how digestion of fat is affected by the process related to the components (1) x2 (4)
- e.g.
blockage of duct B by gallstones will result in: (any **two** of the following sets)
• decrease in the secretion of bile salts (1), which in turn decreases the emulsification of fat in the small intestine (1)
• decrease in the secretion of lipase (1), which in turn decreases the chemical digestion of fat in the small intestine (1)
• decrease in the secretion of sodium hydrogen carbonate (1) such that the pH in the small intestine is no longer optimum for the digestion of fat (1)

5 marks

2022Q1

- 1 (a) D = C - (any combination) (1) (1)
- (b) Concept for mark award:
• vegetarian diets contain a large amount of dietary fibre which is indigestible (1)
• the indigestible materials will be egested as faeces (1)
• impact on D/C ratio (1) (3)
- e.g.
• vegetarians feed on plant materials, a large proportion of which is indigestible / contains dietary fibre / cellulose which is indigestible (1)
• these indigestible materials cannot be absorbed and will be egested as faeces (1)
• as a result, the vegetarians have a greater energy loss through faeces / excretion than non-vegetarians (1), leading to a lower D/C ratio

4 marks

2022Q11

11 Concept for mark award:

Managing the blood glucose level in diabetics (max. 3):

- as compared with normal diets, the consumption of ketogenic diets will not lead to a sharp rise in blood glucose level / result in a relatively stable blood glucose (1) because
 - the ketogenic diet is low in carbohydrates, this will end up with a smaller amount of glucose after digestion and absorption (1)
 - after metabolism, only a small amount of glucose will be obtained from the fat of the ketogenic diet (1)
- diabetics cannot adequately regulate the blood glucose level on their own / cannot produce sufficient insulin to lower the blood glucose level after meal / cannot adequately respond to insulin (1), after consumption of ketogenic diet, the blood glucose level is still manageable to the diabetics / glucose will not appear in the urine of the diabetics (1)

max. 3

Concept of energy balance for weight management (max. 4):

- to lose weight, the daily energy intake should be lower than the daily energy expenditure / energy deficit (1), when energy intake is lower than energy expenditure, the body will mobilise / use the energy reserve, such as glycogen stored in liver and muscles as energy source (1)
- ketogenic diet is low in carbohydrates, the body is forced to use stored glycogen as energy source (1), if glycogen is insufficient / depleted, the body will use the stored fat instead (1)
- as the ketogenic diet has a high fat content, the energy intake may be higher than the normal diet (1) because fat stores more energy than the same mass of proteins and carbohydrates (1)

max. 4

Other health concerns (max. 2): (Any **two**, accept other reasonable answers)

- the high-fat content may result in obesity or blockage of blood vessels due to high fat / triglyceride / cholesterol levels in blood which may result in cardiovascular diseases / stroke (1)
- brain cells use mainly glucose as the energy source, a low carbohydrate intake may result in a low blood glucose supply, leading to dizziness (1)
- consuming a ketogenic diet will deplete the glycogen store, people may feel tired easily (1)

max. 2

Effective communication (0-3)

max. 3
12 marks

2023Q10a, bi, bii1, bii2

- 10 (a) • mouth cavity / small intestine (1)
- (b) (i) $11100 / 2675 \times 250 = 1037.5 \text{ g}$ (1)
[no unit no mark]
- (ii) (1) $11100 / 2675 \times 3.5 = 14.5 \text{ g}$ (1)
[no unit no mark]
- (2) • his blood protein level will be lower than normal people (1)
• because his protein intake is much lower than the recommended daily requirement (1) (2)

2024Q6

	Concept for mark award	Example	Marks
6(a)	correct choice	pancreatic amylase	(1)
	explanation	most of the starch digestion takes place in the small intestine / small intestine is long which allows more time for starch digestion	(1)
		while pancreatic amylase is responsible for the digestion of starch in the small intestine	(1)
6(b)(i)	concept of control variables	to ensure the volume of the reaction mixture in the two set-ups is the same	(1)
		to ensure that the concentration of substrate / enzyme in the two set-ups is the same because it will affect the rate of digestion	(1)
6(b)(ii)	correct reagent / test for detecting the substrate / product + valid measurement of the change in substrate / product over time	either one of the following sets: - use iodine test / iodine solution - add a drop of the reacting mixture to iodine solution at fixed time intervals, then measure the time taken for blue-black color to disappear / for a negative result from the test Or - use Benedict's test / Benedict's solution - at a fixed time interval, take a fixed amount of the reacting mixture and heat with Benedict's solution, measure the amount of brick-red precipitate formed / measure the absorbance / transmittance / optical density of the supernatant	(2)
6(c)	effect of inhibitor on enzyme activity	same curve before time t, after that the curve is lower than the original curve	(1)
6(d)(i)	validity of the results	any one of the following: - the conditions in the human body / animal's body (in vivo) are a lot more complicated than reactions in a test tube (in vitro) - there are multiple factors affecting the reaction in the body - reaction in a test tube cannot simulate the conditions in a body	(1)
6(d)(ii)	correct blood composition	blood glucose level	(1)
	correct results related to starch digestion	a slower rise in blood glucose level in the experimental group than the control group / lower blood glucose level in the experimental group than the control group	(1)
6(e)	inhibition of starch digestion alters the food preference of insects	insects cannot digest the starch / cannot obtain nutrients from the beans with the amylase inhibitor, in the long run, insects will feed on other plants for food / insects will not eat the beans with amylase inhibitor	(1)

12 marks